Low-Level Design: Bill Types

Date: 2026-06-13 | **Plan**: plan.md | **HLD**: hld.md

Data Model Changes

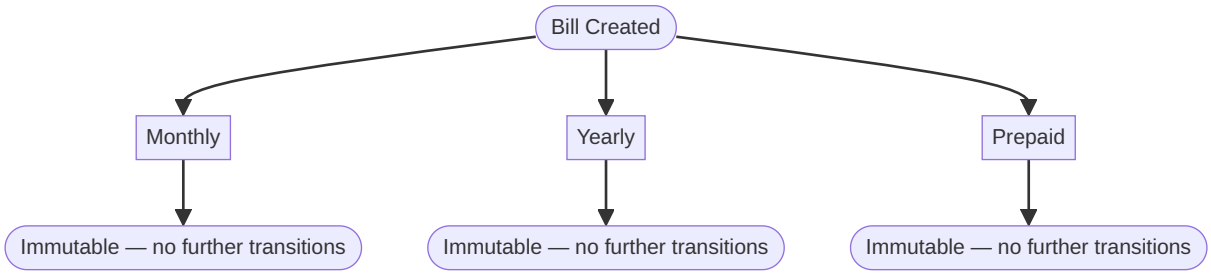
No new columns. `BillingPeriod` already exists in the `bills` table.

BILL		
int	Id	PK
string	Name	
string	BillingPeriod	
int	BillingDate	
int	PayByDate	
int	LatePayByDate	
boolean	IsEnabled	
string	PrimaryType	
date	CreatedAt	
date	UpdatedAt	

> `BillingPeriod` allowed values: `monthly`, `yearly`, `prepaid` — immutable after creation. `BillingDate`, `PayByDate`, and `LatePayByDate` are not required for prepaid bills.

Validation rules: - `BillingPeriod` MUST be one of: `monthly`, `yearly`, `prepaid`. - If omitted on create, defaults to `"monthly"`. - Write-once: set at creation and never updated thereafter. - When `prepaid`: `BillingDate`, `PayByDate`, and `LatePayByDate` are optional.

Billing type lifecycle:



DTO Changes

`bills` service

`CreateBillDTO` — add field: - `BillingPeriod` (string, default `"monthly"`)

`Bill` model — no change (`BillingPeriod` already present).

`BillDTO` — no change (`BillingPeriod` already present and mapped).

`BillsMappingProfile` — no explicit mapping needed. AutoMapper resolves `BillingPeriod` by name convention from `CreateBillDTO` to `Bill`. Verify in tests.

Frontend Changes

Add Bill page (`routes/bills/add/+page.svelte`)

What changes: Add billing type selection to the add bill form.

Component decision: - Build a new `BillingTypePicker` inline component (three pill/toggle buttons: **monthly** | **yearly** | **prepaid**). - Default selection: `monthly`. - Wire selected value into the `addBill` mutation as `billingPeriod`. - When `prepaid` is selected, hide the `BillingDate` and `PayByDate` inputs (they are not required).

Bills list and detail page (`routes/bills/+page.svelte`)

What changes: Display billing type on each bill in the list and in the detail/edit view.

Component decision: - Build a new `BillingTypeBadge` inline component (read-only pill/badge). - Reuse for both the list row and the detail view — no edit control rendered. - Add `billingPeriod` to the `bills` query and `billById` query selection sets. - Fallback: if `billingPeriod` is null or empty, display `"monthly"`.